Physics of Complex Systems

Lecture and Tutorials – Prof. Dr. Haye Hinrichsen – Maximilian Schäfferer – SS 2026

```
In[1]:= Attributes[CircleTimes] = {Flat, OneIdentity};
        CircleTimes[a_List /; VectorQ[a], b_List /; VectorQ[b]] :=
          Flatten[KroneckerProduct[a, b]];
        CircleTimes[a_List /; MatrixQ[a], b_List /; MatrixQ[b]] :=
          KroneckerProduct[a, b];

In[4]:= {a, b} ⊗ {c, d, e}

Out[4]= {a c, a d, a e, b c, b d, b e}

In[5]:= sx = {{0, 1}, {1, 0}};
        sx ⊗ sx

Out[6]= {{0, 0, 0, 1}, {0, 0, 1, 0}, {0, 1, 0, 0}, {1, 0, 0, 0}}
```

Automated tensor product in Mathematica

Exercise 5.1: Tensor product (6P)

Let $\mathbf{A}$ and $\mathbf{B}$ be square matrices acting on the vector space V.

(a) Prove that $e^{\mathbf{A}\otimes\mathbb{1}+\mathbb{1}\otimes\mathbf{B}} = e^{\mathbf{A}} \otimes e^{\mathbf{B}}$. (2P)

(b) Consider the eigenvectors and eigenvalues of $\mathbf{A}|a\rangle = a|a\rangle$ and $\mathbf{B}|b\rangle = b|b\rangle$. What are the eigenvalues and eigenvectors of $\mathbf{A}\otimes\mathbf{B}$? (1P)

(c) Compute the commutators of the Pauli matrices $[\sigma_i, \sigma_j]$ and $[\sigma_i \otimes \sigma_i, \sigma_j \otimes \sigma_j]$. (1P)

(d) Compute $e^{\phi\, \sigma_i \otimes \sigma_j}$ for $\phi \in \mathbb{R}$. (2P)

Exercise 5.2: Creation-Annihilation Model (6P)

Consider a one-dimensional chain of three sites which are either empty ($\emptyset$) or occupied by a single particle (A), assuming closed (non-periodic) boundary conditions. On neighboring sites the particles diffuse and react with each other according to the following rules:

$\emptyset A \leftrightarrow A\emptyset$	diffusion at rate D
$\emptyset A / A\emptyset \to AA$	offspring production at rate λ
$AA \to \emptyset\emptyset$	pair annihilation ar rate μ.

(a) Construct the 4×4 interaction matrix of the two-site Liouville operator $\mathcal{L}^{(2)}$ in the canonical configuration basis $(\emptyset\emptyset, \emptyset A, A\emptyset, AA)$. (2P)

(b) Determine the 8×8 matrix of the Liouville-Operator

$$\mathcal{L}^{(3)} = \mathbb{1} \otimes \mathcal{L}^{(2)} + \mathcal{L}^{(2)} \otimes \mathbb{1}$$

acting on a chain with three sites ($\mathbb{1}$ denotes the 2×2 unit matrix). (2P)

(c) Consider the special case $\lambda = 0$ and calculate the eigenvalues of $\mathcal{L}^{(3)}$. Why is the ground state two-fold degenerate? (2P)

(Σ = 12P)

Please submit your solution electronically as a single PDF file until Wednesday, May 20, 2026, via Wue-Campus.